EXPERIMENT NO 8

8.1.1 Multiplication of two square matrices

Algorithm

Step 1: Start the program.

Step 2: Input the dimension n of the square matrices.

Step 3: Input the elements of the first matrix A of size n × n.

Step 4: Input the elements of the second matrix B of size n × n.

Step 5: Initialize i = 0.

Step 6: Check if i < n.

- If **No**, go to **Step 18**.
- If **Yes**, go to **Step 7**.

Step 7: Initialize j = 0.

Step 8: Check if j < n.

- If **No**, go to **Step 16**.
- If **Yes**, go to **Step 9**.

Step 9: Set C[i][j] = 0.

Step 10: Initialize k = 0.

Step 11: Check if k < n.

- If **No**, go to **Step 14**.
- If **Yes**, go to **Step 12**.

Step 12: Compute
C[i][j] = C[i][j] + A[i][k] × B[k][j].

Step 13: Increment k = k + 1 and go back to **Step 11**.

Step 14: Increment j = j + 1.

Step 15: Go back to **Step 8**.

Step 16: Increment i = i + 1.

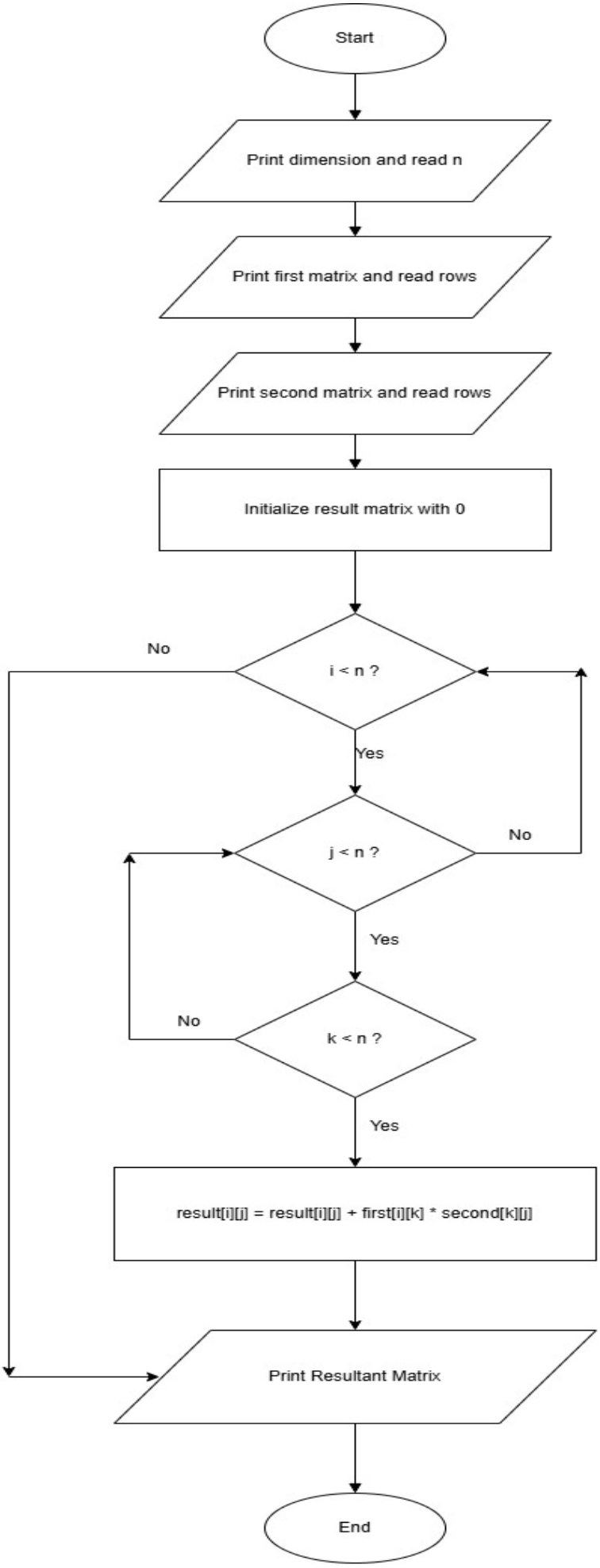
Step 17: Go back to **Step 6**.

Step 18: Print the resultant matrix C.

Step 19: End the program.

Flowchart

EXPERIMENT NO 8

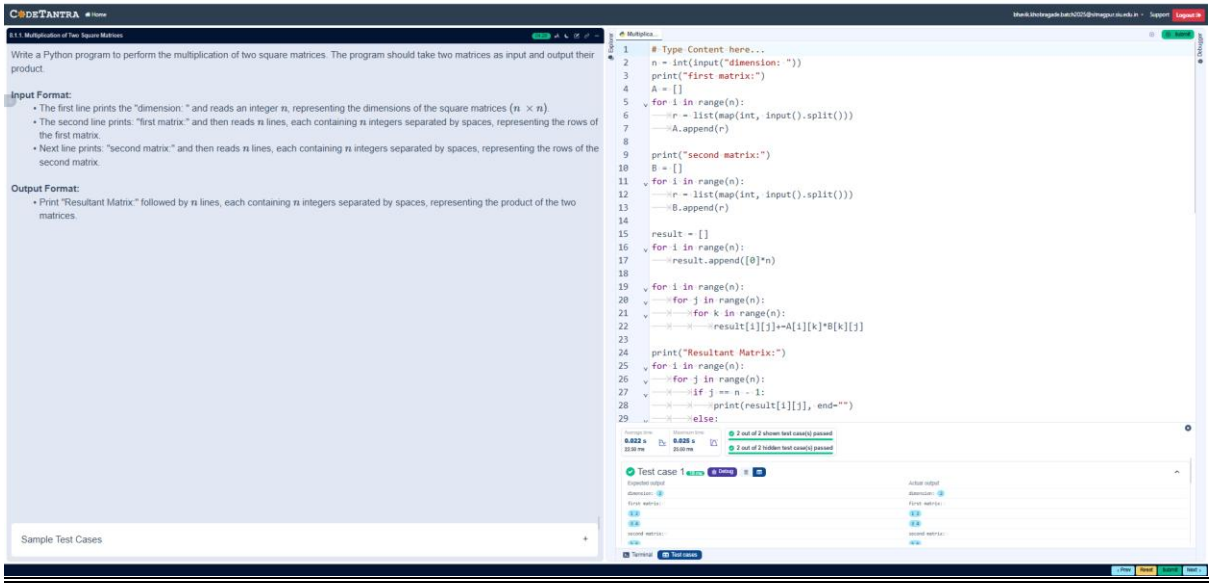


EXPERIMENT NO 8

PYTHON CODE

```
n = int(input("dimension: "))
print("first matrix:")
A = []
for i in range(n):
    row = list(map(int, input().split()))
    A.append(row)
print("second matrix:")
B = []
for i in range(n):
 row = list(map(int, input().split()))
 B.append(row)
result = []
for i in range(n):
    result.append([0] * n)
for i in range(n):
    for j in range(n):
        for k in range(n):
            result[i][j] += A[i][k] * B[k][j]


print("Resultant Matrix:")
for row in result:
    print(*row)
```

EXCECUTION

EXPERIMENT NO 8